INCH-POUND

A-A-60040

19 September 2022

COMMERCIAL ITEM DESCRIPTION

CLAMPS, CADMIUM AND HEXAVALENT-CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers clamps which shall not contain cadmium or hexavalent-chromium in their finishes.

2 CLASSIFICATION

2.1 Types.

Type I	-	NASM21322 - Clamp, Loop, Aluminum, Plain and Cushioned
Type II	-	NASM21333 - Clamp, Loop, Steel, Plain and Cushioned
Type III	-	NASM21334 - Clamp, Loop, Double, Cable Or Tube, Steel, Plain and Cushioned

3 SALIENT CHARACTERISTICS.

3.1 Materials. Clamps shall be manufactured as specified in the Aerospace Industry Association (AIA) documents listed in 2.1 by Type, except as noted, and the plating/finishes shall not contain cadmium and hexavalent-chromium as specified herein.

3.2 Dimensions. Dimensions for all designated types of clamps apply after plating/finishing. Refer to each individual NAS/NASM specification listed in 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent-chromium free part identification codes (PIN) and finish/plating options. Finish/plating options for each type of grommet shall be specified in the contract. PIN codes and finish/plating options for all designated as specified in the contract for the grommets are:

J	=	Aluminum clamps shall be chemically treated in accordance with MIL-DTL-5541, Type II. Sealers (clear or any color) containing hexavalent chromium shall be prohibited.
K	-	Anodized in accordance with MIL-PRF-8625, Type IC or IIB, Class 1, Sealers (clear or any color) containing hexavalent chromium shall be prohibited.
ZN	=	Carbon and alloy steel clamps shall be Zn-Ni plated in accordance with SAE AMS2461, Grade A, Class 2 or ASTM F1941/F1941M Fe/Zn-Ni 8BNS.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 5340

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3.4 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent-chromium with the cadmium and hexavalent-chromium free parts listed herein.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 13834; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections.

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packaging, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Table. The following Tables indicate new finish/plating PIN's which are cadmium and hexavalent-chromium free for the designated clamps.

TABLE I – Type I (-1) - Clamp, loop, aluminum, plain and cushioned.

A-A-60040 Plain Basic Part Number	Reference Dash Number Plain (from NASM21322)
AA60040-1-1*	MS21322-1
AA60040-1-2*	MS21322-2
AA60040-1-3*	MS21322-3
AA60040-1-4*	MS21322-4
AA60040-1-5*	MS21322-5
AA60040-1-6*	MS21322-6
AA60040-1-7*	MS21322-7

A-A-60040 Cushioned Basic Part Number	Reference Dash Number Cushioned (from NASM21322)
AA60040-1-32*	MS21322-32
AA60040-1-33*	MS21322-33
AA60040-1-34*	MS21322-34
AA60040-1-35*	MS21322-35
AA60040-1-36*	MS21322-36
AA60040-1-37*	MS21322-37
AA60040-1-38*	MS21322-38

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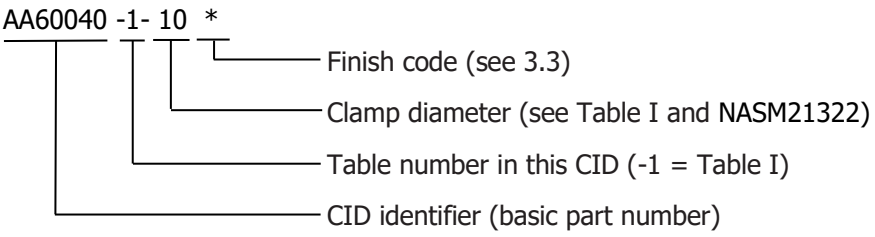
TABLE I – Type I (-1) - Clamp, loop, aluminum, plain and cushioned - Continued.

AA60040-1-8*	MS21322-8	AA60040-1-39*	MS21322-39
AA60040-1-9*	MS21322-9	AA60040-1-40*	MS21322-40
AA60040-1-10*	MS21322-10	AA60040-1-41*	MS21322-41
AA60040-1-11*	MS21322-11	AA60040-1-42*	MS21322-42
AA60040-1-12*	MS21322-12	AA60040-1-43*	MS21322-43
AA60040-1-13*	MS21322-13	AA60040-1-44*	MS21322-44
AA60040-1-14*	MS21322-14	AA60040-1-45*	MS21322-45
AA60040-1-15*	MS21322-15	AA60040-1-46*	MS21322-46
AA60040-1-16*	MS21322-16	AA60040-1-47*	MS21322-47
AA60040-1-17*	MS21322-17	AA60040-1-48*	MS21322-48
AA60040-1-18*	MS21322-18	AA60040-1-49*	MS21322-49
AA60040-1-19*	MS21322-19	AA60040-1-50*	MS21322-50
AA60040-1-20*	MS21322-20	AA60040-1-51*	MS21322-51
AA60040-1-21*	MS21322-21	AA60040-1-52*	MS21322-52
AA60040-1-22*	MS21322-22	AA60040-1-53*	MS21322-53
AA60040-1-23*	MS21322-23	AA60040-1-54*	MS21322-54
AA60040-1-24*	MS21322-24	AA60040-1-55*	MS21322-55
AA60040-1-25*	MS21322-25	AA60040-1-56*	MS21322-56
AA60040-1-26*	MS21322-26	AA60040-1-57*	MS21322-57
AA60040-1-27*	MS21322-27	AA60040-1-58*	MS21322-58
AA60040-1-28*	MS21322-28	AA60040-1-59*	MS21322-59
AA60040-1-29*	MS21322-29	AA60040-1-60*	MS21322-60
AA60040-1-30*	MS21322-30	AA60040-1-61*	MS21322-61
AA60040-1-31*	MS21322-31	AA60040-1-62*	MS21322-62

* Designates finish: J = Chemically treated; K = Anodized (see 3.3).

NOTES

PIN construction



Examples of part number:

AA60040-1-10J	=	a NASM21322 - Clamp, plain, 0.688 nominal dia., aluminum, chemically treated.
AA60040-1-10K	=	a NASM21322 - Clamp, plain, 0.688 nominal dia., aluminum, anodized.
AA60040-1-35J	=	a NASM21322 - Clamp, cushioned, 0.312 nominal dia., aluminum, chemically treated.
AA60040-1-35K	=	a NASM21322 - Clamp, cushioned, 0.312 nominal dia., aluminum, anodized.

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TABLE II – Type II (-2) - Clamp, loop, steel, plain and cushioned.

A-A-60040 Plain Basic Part Number	Reference Dash Number Plain (from NASM21333)	A-A-60040 Cushioned* Basic Part Number	Reference Dash Number Cushioned (from NASM21333)
AA60040-2-1ZN	MS21333-1	AA60040-2-65*ZN	MS21333-65
AA60040-2-2ZN	MS21333-2	AA60040-2-66*ZN	MS21333-66
AA60040-2-3ZN	MS21333-3	AA60040-2-67*ZN	MS21333-67
AA60040-2-4ZN	MS21333-4	AA60040-2-68*ZN	MS21333-68
AA60040-2-5ZN	MS21333-5	AA60040-2-69*ZN	MS21333-69
AA60040-2-6ZN	MS21333-6	AA60040-2-70*ZN	MS21333-70
AA60040-2-7ZN	MS21333-7	AA60040-2-71*ZN	MS21333-71
AA60040-2-8ZN	MS21333-8	AA60040-2-72*ZN	MS21333-72
AA60040-2-9ZN	MS21333-9	AA60040-2-73*ZN	MS21333-73
AA60040-2-10ZN	MS21333-10	AA60040-2-74*ZN	MS21333-74
AA60040-2-11ZN	MS21333-11	AA60040-2-75*ZN	MS21333-75
AA60040-2-12ZN	MS21333-12	AA60040-2-76*ZN	MS21333-76
AA60040-2-13ZN	MS21333-13	AA60040-2-77*ZN	MS21333-77
AA60040-2-14ZN	MS21333-14	AA60040-2-78*ZN	MS21333-78
AA60040-2-15ZN	MS21333-15	AA60040-2-79*ZN	MS21333-79
AA60040-2-16ZN	MS21333-16	AA60040-2-80*ZN	MS21333-80
AA60040-2-17ZN	MS21333-17	AA60040-2-81*ZN	MS21333-81
AA60040-2-18ZN	MS21333-18	AA60040-2-82*ZN	MS21333-82
AA60040-2-19ZN	MS21333-19	AA60040-2-83*ZN	MS21333-83
AA60040-2-20ZN	MS21333-20	AA60040-2-84*ZN	MS21333-84
AA60040-2-21ZN	MS21333-21	AA60040-2-85*ZN	MS21333-85
AA60040-2-22ZN	MS21333-22	AA60040-2-86*ZN	MS21333-86
AA60040-2-23ZN	MS21333-23	AA60040-2-87*ZN	MS21333-87
AA60040-2-24ZN	MS21333-24	AA60040-2-88*ZN	MS21333-88
AA60040-2-25ZN	MS21333-25	AA60040-2-89*ZN	MS21333-89
AA60040-2-26ZN	MS21333-26	AA60040-2-90*ZN	MS21333-90
AA60040-2-27ZN	MS21333-27	AA60040-2-91*ZN	MS21333-91
AA60040-2-28ZN	MS21333-28	AA60040-2-92*ZN	MS21333-92
AA60040-2-29ZN	MS21333-29	AA60040-2-93*ZN	MS21333-93
AA60040-2-30ZN	MS21333-30	AA60040-2-94*ZN	MS21333-94
AA60040-2-31ZN	MS21333-31	AA60040-2-95*ZN	MS21333-95
AA60040-2-32ZN	MS21333-32	AA60040-2-96*ZN	MS21333-96
AA60040-2-33ZN	MS21333-33	AA60040-2-97*ZN	MS21333-97
AA60040-2-34ZN	MS21333-34	AA60040-2-98*ZN	MS21333-98
AA60040-2-35ZN	MS21333-35	AA60040-2-99*ZN	MS21333-99
AA60040-2-36ZN	MS21333-36	AA60040-2-100*ZN	MS21333-100
AA60040-2-37ZN	MS21333-37	AA60040-2-101*ZN	MS21333-101
AA60040-2-38ZN	MS21333-38	AA60040-2-102*ZN	MS21333-102
AA60040-2-39ZN	MS21333-39	AA60040-2-103*ZN	MS21333-103
AA60040-2-40ZN	MS21333-40	AA60040-2-104*ZN	MS21333-104
AA60040-2-41ZN	MS21333-41	AA60040-2-105*ZN	MS21333-105
AA60040-2-42ZN	MS21333-42	AA60040-2-106*ZN	MS21333-106
AA60040-2-43ZN	MS21333-43	AA60040-2-107*ZN	MS21333-107
AA60040-2-44ZN	MS21333-44	AA60040-2-108*ZN	MS21333-108

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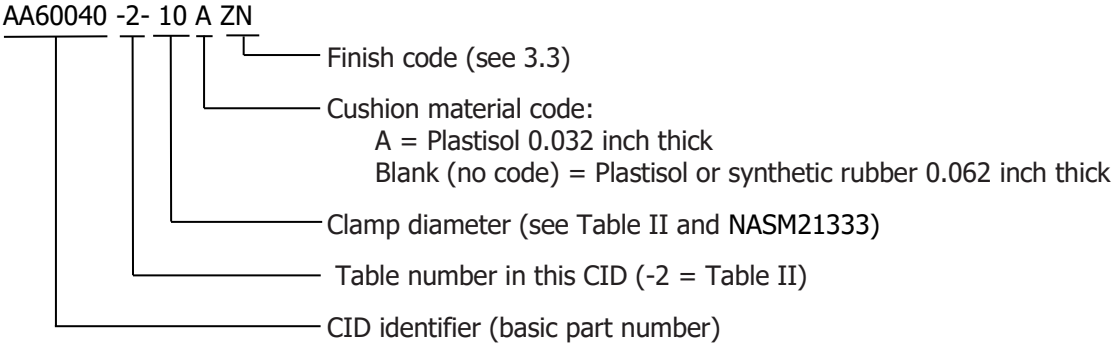
TABLE II – Type II (-2) - Clamp, loop, steel, plain and cushioned – Continued.

AA60040-2-45ZN	MS21333-45	AA60040-2-109*ZN	MS21333-109
AA60040-2-46ZN	MS21333-46	AA60040-2-110*ZN	MS21333-110
AA60040-2-47ZN	MS21333-47	AA60040-2-111*ZN	MS21333-111
AA60040-2-48ZN	MS21333-48	AA60040-2-112*ZN	MS21333-112
AA60040-2-49ZN	MS21333-49	AA60040-2-113*ZN	MS21333-113
AA60040-2-50ZN	MS21333-50	AA60040-2-114*ZN	MS21333-114
AA60040-2-51ZN	MS21333-51	AA60040-2-115*ZN	MS21333-115
AA60040-2-52ZN	MS21333-52	AA60040-2-116*ZN	MS21333-116
AA60040-2-53ZN	MS21333-53	AA60040-2-117*ZN	MS21333-117
AA60040-2-54ZN	MS21333-54	AA60040-2-118*ZN	MS21333-118
AA60040-2-55ZN	MS21333-55	AA60040-2-119*ZN	MS21333-119
AA60040-2-56ZN	MS21333-56	AA60040-2-120*ZN	MS21333-120
AA60040-2-57ZN	MS21333-57	AA60040-2-121*ZN	MS21333-121
AA60040-2-58ZN	MS21333-58	AA60040-2-122*ZN	MS21333-122
AA60040-2-59ZN	MS21333-59	AA60040-2-123*ZN	MS21333-123
AA60040-2-60ZN	MS21333-60	AA60040-2-124*ZN	MS21333-124
AA60040-2-61ZN	MS21333-61	AA60040-2-125*ZN	MS21333-125
AA60040-2-62ZN	MS21333-62	AA60040-2-126*ZN	MS21333-126
AA60040-2-63ZN	MS21333-63	AA60040-2-127*ZN	MS21333-127
AA60040-2-64ZN	MS21333-64	AA60040-2-128*ZN	MS21333-128
		AA60040-2-129*ZN	MS21333-129

* Cushion material code (if any).

NOTES (see next page)

PIN construction



Examples of part number:

AA60040-2-10ZN = a NASM21333 - Loop clamp, 0.688-inch nominal dia., plain, carbon steel, plastisol or synthetic rubber cushion 0.062-inch thick, overall width of 0.500-inch, Zinc-Nickel plated.

AA60040-2-66AZN = a NASM21333 – Loop clamp, 0.188-inch nominal dia., carbon steel, cushioned, plastisol cushion 0.032-inch thick, overall width of 0.625-inch, Zinc-Nickel plated.

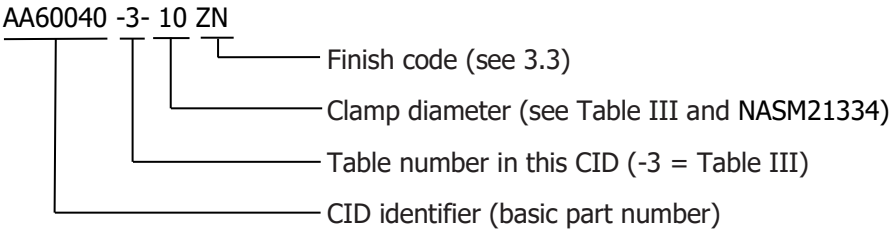
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TABLE III – Type III (-3) - Clamp, loop - double, cable or tube, steel, plain and cushioned.

A-A-60040 Plain Basic Part Number	Reference Dash Number Plain (from NASM21334)		A-A-60040 Cushioned Basic Part Number	Reference Dash Number Cushioned (from NASM21334)
AA60040-3-1ZN	MS21334-1		AA60040-3-22ZN	MS21334-22
AA60040-3-2ZN	MS21334-2		AA60040-3-23ZN	MS21334-23
AA60040-3-3ZN	MS21334-3		AA60040-3-24ZN	MS21334-24
AA60040-3-4ZN	MS21334-4		AA60040-3-25ZN	MS21334-25
AA60040-3-5ZN	MS21334-5		AA60040-3-26ZN	MS21334-26
AA60040-3-6ZN	MS21334-6		AA60040-3-27ZN	MS21334-27
AA60040-3-7ZN	MS21334-7		AA60040-3-28ZN	MS21334-28
AA60040-3-8ZN	MS21334-8		AA60040-3-29ZN	MS21334-29
AA60040-3-9ZN	MS21334-9		AA60040-3-30ZN	MS21334-30
AA60040-3-10ZN	MS21334-10		AA60040-3-31ZN	MS21334-31
AA60040-3-11ZN	MS21334-11		AA60040-3-32ZN	MS21334-32
AA60040-3-12ZN	MS21334-12		AA60040-3-33ZN	MS21334-33
AA60040-3-13ZN	MS21334-13		AA60040-3-34ZN	MS21334-34
AA60040-3-14ZN	MS21334-14		AA60040-3-35ZN	MS21334-35
AA60040-3-15ZN	MS21334-15		AA60040-3-36ZN	MS21334-36
AA60040-3-16ZN	MS21334-16		AA60040-3-37ZN	MS21334-37
AA60040-3-17ZN	MS21334-17		AA60040-3-38ZN	MS21334-38
AA60040-3-18ZN	MS21334-18		AA60040-3-39ZN	MS21334-39
AA60040-3-19ZN	MS21334-19		AA60040-3-40ZN	MS21334-40
AA60040-3-20ZN	MS21334-20		AA60040-3-41ZN	MS21334-41
AA60040-3-21ZN	MS21334-21		AA60040-3-42ZN	MS21334-42

NOTES

PIN construction



Examples of part number:

AA60040-3-10ZN = a NASM21334 - Loop clamp, 0.688-inch nominal dia., plain, carbon steel, 0.062-inch thick, overall width of 0.500-inch, Zinc-Nickel plated.

AA60040-3-37ZN = a NASM21334 – Loop clamp, 1.375-inch nominal dia., carbon steel, cushioned, plastisol or synthetic rubber cushion 0.094-inch thick, overall width of 0.625-inch, Zinc-Nickel plated.

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7.2 Source of documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/browse/index/far.

7.2.2 Executive order (EO). Copies of the EO documents are available online from https://www.epa.gov/greeningepa/executive-order-13834.

7.2.3 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.4 Non-Government specifications: Copies of Aerospace Industry Association (AIA) publications are available online from https://www.aia-aerospace.org/standards/. Copies of American Society for Testing and Materials (ASTM) are available online from https://www.astm.org/. Copies of Society of Automotive Engineers (SAE) are available online from https://www.sae.org/standards/.

7.3 Ordering data. The contract or order should specify the following:

a. Title, number and date of this CID.
b. Federal stock number, (if available).
c. Material (steel alloy, steel, or aluminum).
d. Part identification number (PIN).
e. Quantity.

7.4 Key words:

a. Anodize
b. Chemically treated
c. Finish
d. Plating
e. Steel
f. Steel alloy

MILITARY INTERESTS:

Custodian:
Army – AT
Navy – YD
Air Force – 71
DLA – CC

Review activities:
Army – CR, MI
Navy –MC, OS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 5340-2022-004)

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